

MORPHOS: Toward Matter That Builds and Heals Itself

A first-principles case for morphogenetic fabrication

Moonshot Hackathon (Aethra) — prototype, paper, and vision.

Abstract. Human manufacturing rests on a hidden assumption: that we make things by **assembling** parts according to a master plan. Biology rejects that assumption. A living body is not assembled; it **grows** from a single cell following purely local rules — and, crucially, it **repairs itself** when damaged. MORPHOS is a prototype and a thesis arguing that the next era of fabrication, materials, and robotics will move from *assembly* to *morphogenesis*: structures whose form, function, and resilience emerge from simple rules running locally in every unit of matter. We demonstrate the core mechanism with a Neural Cellular Automaton (NCA) that grows a target structure from one seed cell and autonomously regenerates after arbitrary damage, and we lay out the path from pixels to programmable physical matter. We are explicit about prior art: the NCA mechanism and reaction–diffusion morphogenesis are published work. Our contribution is the **synthesis, the fabrication thesis, and an interactive self-repair design tool.**

1. The problem we have misunderstood

We treat robustness as something you **add** to a manufactured object — redundancy, maintenance, inspection, replacement. But the most robust systems we know were never *manufactured* in the assembly sense at all. A salamander regrows a lost limb. A forest reseeds a burned clearing. A cut heals. None of these systems contains a central controller issuing instructions; each is a population of simple agents (cells) following local rules, with global order as an emergent consequence.

The misunderstanding is this: **we have conflated "designing a structure" with "specifying every part and its placement."** That conflation makes our artifacts brittle. A bridge cannot reroute load around a crack. A chip cannot route around a dead transistor. A building cannot reseal a breach. Each is a frozen blueprint, and a blueprint has no answer to damage except external repair.

2. Why existing solutions are insufficient

The dominant responses to brittleness — redundancy, modular replacement, predictive maintenance, and self-healing chemistries — are all valuable, but they are **patches on the**

assembly paradigm, not alternatives to it. Redundancy multiplies cost and weight. Modular replacement still requires a human or robot to perform the swap. Self-healing polymers reseal micro-cracks but cannot **regrow lost structure** or change function. None of them gives matter the property biology takes for granted: the ability to **regenerate form from local information alone**.

What is missing is not a better material or a better controller. What is missing is a **different organizing principle** — one where the "instructions" live identically inside every unit of matter, so that any surviving fragment contains everything needed to rebuild the whole.

3. The first-principles insight

Alan Turing showed in 1952 that complex, stable spatial patterns can arise from simple local chemical interactions — morphogenesis without a blueprint. MORPHOS builds on the modern, computational form of this idea:

*If the update rule of every cell is identical and depends only on its local neighborhood, then the global shape is an **attractor** of that rule.*

An attractor is, by definition, self-correcting: perturb the system (damage it) and it flows back toward the attractor (it heals). This reframes the entire act of design. To "design a structure" in the morphogenetic paradigm is not to specify a blueprint but to **specify a local rule whose attractor is the structure you want**. The structure then becomes a *dynamic equilibrium* rather than a static arrangement — and dynamic equilibria repair themselves for free.

4. The prototype and its scientific foundations

MORPHOS implements this with a **Neural Cellular Automaton**, building directly on *Growing Neural Cellular Automata* (Mordvintsev, Randazzo, Niklasson, Levin; *Distill*, 2020) and the Turing / Gray–Scott lineage of reaction–diffusion morphogenesis. We state this prior art plainly; our contribution is the synthesis, the thesis, and the interactive tool.

State. The world is a 64×64 grid. Each cell carries a 16-dimensional state: 4 *visible* channels (RGBA) and 12 *hidden* signalling channels — the computational analogue of morphogens in developmental biology.

Perception. Each cell perceives its neighborhood through three fixed depthwise filters applied per channel — an identity filter (itself) and Sobel-x / Sobel-y filters (local gradients). This yields a 48-number perception vector per cell. *No cell ever sees the whole grid.*

Update rule (the only learned part). A tiny shared network — a two-layer 1×1 convolution, $48 \rightarrow 128 \rightarrow 16$ — maps each cell's perception to an incremental **update** of its own state. The final layer is **zero-initialized**, so the organism begins as a literal "do-nothing" rule and must *learn* to act.

Stochastic, asynchronous cells. Only ~50% of cells update on any step (a random per-cell mask). Real cells share no global clock; this asynchrony also makes the learned rule more robust.

Alive masking. A cell is "alive" only if it or a neighbor has $\alpha > 0.1$; dead cells are zeroed. Growth stays anchored to the body — matter cannot teleport into empty space.

The decisive choice — a regeneration curriculum. We maintain a **pool** of past organism states. Each step we sample a batch, reset the worst sample to a fresh seed (so growth-from-scratch is always being learned), and — after warm-up — **damage** half the batch by deleting random disks before simulating 64–96 steps forward. The result is matched to the target RGBA (MSE), with per-parameter gradient normalization (Adam, $\text{lr } 2\text{e-}3$, step decay). Because damaged samples re-enter the loss, **the only way to lower error is to discover update rules that rebuild missing structure from surviving cells.** Self-repair is not hand-coded; it emerges from training a local rule to be a robust attractor.



Figure 1. Growth, damage, and self-repair from the trained organism (frames from `--demo`). Animation: `morphos_demo.gif`.

Result. The trained organism exhibits the three behaviors that are the minimal signature of morphogenetic matter: **(1) growth** from a single seed into the target; **(2)**

regeneration of the correct missing geometry after arbitrary damage; and **(3) homeostasis** — it holds its form when complete, and under a *continuous* stream of small damage it maintains that form indefinitely (a "living material," demonstrated by `--homeostasis`).

5. What changes if this succeeds

Carry the principle from pixels to physics and the implications compound across domains:

- **Infrastructure.** Units that carry the local rule could reroute load around damage and reseal breaches without external repair — bridges, habitats, and pressure vessels that absorb damage instead of failing.
- **Materials.** "Programmable matter" whose macroscopic shape is an attractor of a local rule — reconfigurable on command and self-restoring after wear.
- **Robotics.** Modular and swarm robots that self-assemble into task-specific morphologies and **regrow** function after losing units — resilience by construction, not redundancy.
- **Space & extreme environments.** Where resupply and human repair are impossible, self-healing matter is not a convenience but an enabling condition.
- **Medicine & tissue engineering.** The same mathematics of morphogenetic attractors informs how engineered tissues might be guided to grow and self-correct.

The throughline: **resilience stops being a cost center and becomes a property of the substrate.**

6. The path from concept to reality

MORPHOS is a simulation today; the route to physical matter is a ladder, not a leap:

- **Simulation (done).** A local rule can encode growth, regeneration, and homeostasis. Extend to multi-target (reconfigurable) and 3D rules.
- **Soft / modular robotics.** Map the local rule onto physical modules that sense neighbors and actuate — the first embodiment where "cells" are real.
- **Active materials.** Encode local rules into responsive polymers and micro-robotic colloids, targeting self-healing at the material level.
- **Co-design tools.** Generalize MORPHOS so an engineer specifies a form and the system *learns the rule whose attractor is that form* — designing the rule, not the

blueprint.

Each rung is a tractable research program with existing communities and hardware; none requires new physics.

7. Conclusion

We do not build the way nature builds, and our artifacts are brittle because of it. MORPHOS argues — and demonstrates in miniature — that fabrication can be reconceived as morphogenesis: form as the self-correcting attractor of a local rule, identical in every unit of matter, so that any fragment can rebuild the whole. The prototype grows and heals on a screen today. The thesis is that the same principle, embodied in physical matter, is how we will build things that do not break the way our things break now.

Contributions — what is and isn't ours

Not novel (and we say so): the NCA mechanism (Mordvintsev et al., *Distill* 2020); reaction–diffusion / Turing morphogenesis (Turing 1952; Gray–Scott).

Our zero-to-one contribution: (1) a **fabrication thesis** reframing manufacturing, materials, and robotics around morphogenetic self-repair (assembly→morphogenesis; blueprint→attractor); (2) a **cross-domain map** of what changes when resilience becomes a property of the substrate; (3) an **interactive self-repair simulator / design tool** with a live grow→damage→heal loop and a homeostasis-under-attack mode.

Appendix — Reproducibility

Prototype: `morphos_nca.py` (PyTorch), one file, no dataset required. **Train:** `python morphos_nca.py --train --steps 4000`. **Demo:** `python morphos_nca.py --demo`. **Living material:** `--homeostasis`. **Interactive:** `--interactive`.

Key hyperparameters: 16 channels (4 visible + 12 hidden); 64×64 grid; perception = identity + Sobel-x + Sobel-y; update MLP 48→128→16 (last layer zero-init); fire rate 0.5; pool 1024; batch 8; 64–96 rollout steps; Adam lr 2e-3 (×0.3 at step 2000); per-parameter

gradient normalization; regeneration curriculum (damage half the batch after step 500);
loss = MSE on RGBA; best checkpoint by grow-from-seed validation.

Prior art acknowledged: A. Turing, "The Chemical Basis of Morphogenesis," 1952; reaction–diffusion (Gray–Scott) morphogenesis; A. Mordvintsev, E. Randazzo, E. Niklasson, M. Levin, "Growing Neural Cellular Automata," Distill (2020). MORPHOS contributes the fabrication thesis, the cross-domain framing, and an interactive self-repair simulator/design tool.